

Lecture 10: Content Moderation

HUM 4441: Engineering Ethics for Computer Science Engineers

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AI Recommender Systems

The Chief Means of Content Moderation

Recommender systems **decide what you see**. They are the primary mechanism by which platforms moderate content today - not a separate add-on system.

- Social media recommenders (“algorithms”)
- News media recommenders (“click bait”)
- Retail sites, search engines, GPTs



AI Recommender Systems: Widely Adopted

One of the most **effective** and **widely-adopted** AI applications - a major force in business, marketing, and news media.

Many Successful Techniques

- Matrix factorization
- Bayesian classifiers and decision trees
- Collaborative filtering, k nearest neighbors
- Recurrent neural networks and transformers, etc.

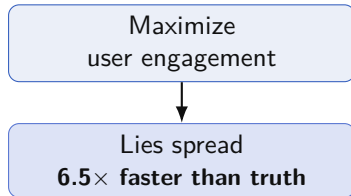


Engagement Maximization: The Downside

AI recommenders are trained to **maximize engagement**. The result is the worldwide spread of:

- **Lies, slander, hate speech, harmful misinformation**
- **Sensational news coverage**

This drives **calls for content moderation** - to avoid harmful side effects of profit-maximizing content algorithms.



Factoid: Lies spread 6.5 times faster on social media than truth.

Forms of Content Moderation

1 Take Down

Removing material entirely from the platform.

2 Demote

Putting material at the end of the recommended list - in effect, the same as taking it down.

3 Flag

Flagging material as false or offensive, often alongside reliable sources.

Recommender systems can implement **all** of these forms of moderation.

Content Moderation: An Ad Hoc Landscape

Current Policies Are Ad Hoc

- Respond to complaints
- Change with ownership and political climate
- We need a **principled approach**

Rather than resolve all the issues, this lecture focuses on **two case studies**: inciting violence on YouTube, and social media's impact on young people.

Note: this is about **voluntary** platform policy, not government regulation.

Case Study 1

Inciting Violence on YouTube

Case Study: Inciting Violence - Background

A very high-profile dilemma that raised issues **not resolved to this day**.

The Film

An amateurish film, *Innocence of Muslims*, was uploaded to YouTube on 1 July 2012.

- Packed with **lies** and misinformation
- Highly **offensive** due to negative portrayal of Islam



Innocence of Muslims

2012 · Drama/Short · 14 mins



Wikipedia

https://en.wikipedia.org/wiki/Innocence_of_Muslims

Innocence of Muslims

Innocence of Muslims is a 2012 anti-Islamic short film written and directed by Nakoula Basseley Nakoula under the pseudonym Sam Bacile. ... Two versions of the 14 ... [Read more](#)

Plot and description

Production

Screening and Internet upload

Case Study: Inciting Violence - Reaction

- Protests worldwide, some **violent**
- Reportedly **50 deaths**, mainly in Pakistan
- Most protests anti-U.S., because the film maker lived in the U.S.
- U.S. government did **not** ban the video



Case Study: Inciting Violence - Google's Response

No Legal Authority

President Barack Obama asked Google (owner of YouTube) to take down the film - but he had **no legal authority** to require it.

Google Refused

Google refused to remove the video globally, but **removed it in some countries.**



Case Study: Inciting Violence - Google's Position

Stated Rationale

The post was consistent with “company policy.”

“It is against the Islam religion [sic] but not Muslim people.”

As reported in the NY Times, 14 Sep 2012.

The Open Question

YouTube now has a laundry list of “community guidelines” - but what is the underlying **principle**?



Generalizability: Is Allowing Lies Deceptive?

- Does merely posting lies cause people to believe something YouTube knows is false?
- Only if users regard the video's appearance as a claim or endorsement **by YouTube**
- But users know YouTube hosts all sorts of contradictory views - so there is no obvious endorsement

Conclusion

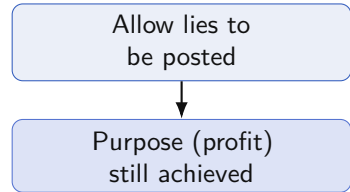
It is hard to make the case that YouTube (as opposed to those who upload the lies) is **deceiving** people.

Generalizability: Would the Policy Still Work?

Would YouTube continue to achieve its purpose even if **all** online sites allowed lies to be posted?

Probably Yes

YouTube's purpose is to **make money**, not to convince anyone to believe its content. Practically all online sites already allow lies, and YouTube still makes tons of money.



Generalizability: Is Removal Generalizable?

Removing False Videos

Removing false videos **because they reduce utility** may **not** be generalizable - generalized private censorship of information may destroy more utility than it creates.

Removing Incendiary Videos

Removing incendiary videos because they **may incite violence** is generalizable - removal would still reduce the probability of violence even as a general practice.

Utilitarian Analysis: Arguments Against Moderation

Argument 1: Causal Distance

"YouTube didn't kill anyone in a riot - the rioters did." **Response:** for the utilitarian principle, only the *consequences* of YouTube's policy matter - it doesn't matter that other people's conduct is involved.

Argument 2: "Ought Implies Can"

Content moderation is a massive task: YouTube gets 500 hours of video/minute; Facebook has 3 billion daily logins. It is **impossible** to screen everything.

Response to Argument 2

- This is a **factual** claim, not an ethical one
- Sites already use AI-based content moderation to direct relevant content to users
- AI can flag questionable posts for human review; users also request takedowns

Scale of Existing Moderation

- YouTube: **10,000** content moderators
- Facebook: **15,000** of 180,000+ employees
- About **100,000** content moderators worldwide

Utilitarian Argument *For* Content Moderation

The Argument

Moderation **increases utility**. Even if it's impossible to catch all harmful posts, it's possible to catch many of them. Failure to do so is a clear violation of the utilitarian principle.

Rebuttal and Counter-rebuttal

"We already do all we can - e.g., we removed all COVID vaccine misinformation."

If so, great - then you agree you **should** do so. And existing recommender technology can clearly be redirected toward this end.

Argument 3: Free Speech

Original Claim

Content moderation violates the **First Amendment rights** of users.

Response: the First Amendment restricts only **government** action - YouTube is a **private company**. This is a generalization argument.

Revised Argument

Content moderation **restricts free speech**, and this is harmful to society.

Question: is it a restriction - or a refusal to give one a particular megaphone?

Autonomy: The Most Demanding Principle

Malicious rumors, terrorist posts, and medical misinformation can lead to death and injury - e.g., in riots. Isn't this a violation of autonomy?

Applying the Principle

A post should be removed when one is **rationally constrained to believe** debilitating harm will result - regardless of who immediately causes the harm.

Implied Consent?

Rioters voluntarily assumed the risk of joining a riot - no autonomy violation for them. But if **innocent bystanders** were hurt, an autonomy violation **does** occur.

Autonomy: How Demanding Can It Get?

Perhaps YouTube managers are rationally constrained to believe that the very **existence** of YouTube will, at some point, interfere with ethical action plans without implied consent - despite best efforts to remove offensive videos.

YouTube's moderation must be strong enough that managers can reasonably believe the platform will not cause an autonomy violation.

So "not irrational" basically means reasonable.

Why is this called the "bar for staying in business"?

Imagine YouTube's managers know:

"No matter what we do, eventually some dangerous video on our platform will cause serious harm to an innocent person."

If they know this is basically inevitable, then under the very strict autonomy principle discussed in the previous slide, simply saying:

"We tried our best to remove dangerous videos"

might not be enough ethically.

The argument becomes:

If operating the platform inevitably violates someone's autonomy, should the platform continue operating at all?

Take Down Misinformation in General?

- **Not required** by the autonomy principle
- **Could** be justified by the utilitarian principle
- But could also be **ungeneralizable**

A Middle-Ground Option

Flag what fact checkers see as misinformation while providing reliable sources. This does not stifle free speech - and may even promote it - while satisfying the utilitarian principle.

Case Study 1: Summing Up

YouTube must shut down unless it adopts a moderation policy for which managers can rationally believe autonomy violations are not inevitable



A strong **utilitarian imperative** exists to identify such a policy, given YouTube's many benefits



A policy of removing false claims must be carefully crafted to avoid violating **generalizability**



Compromise: flag clearly false claims while providing reliable sources

Case Study 2

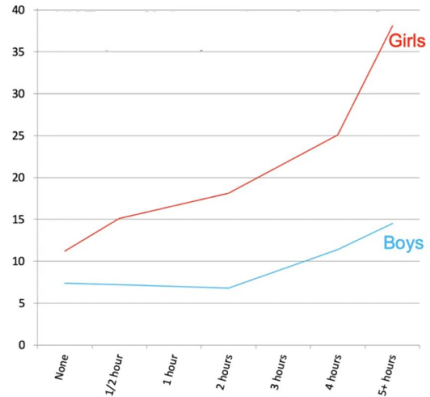
Impact of Social Media on Young People

Impact on Young People: The Data

Depression, Anxiety, and Suicide

All are rising among young people in some countries - roughly coinciding with the rise of social media use.

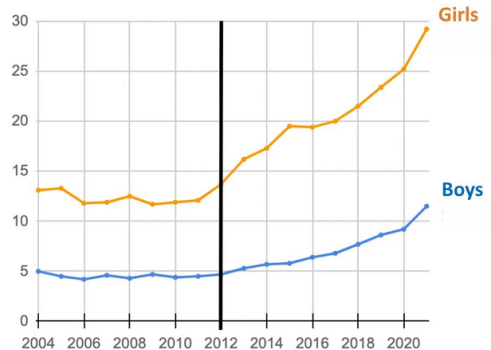
Percent of UK Teens Depressed as a Function of Hours per Weekday on Social Media



Impact on Young People: U.S. Trends

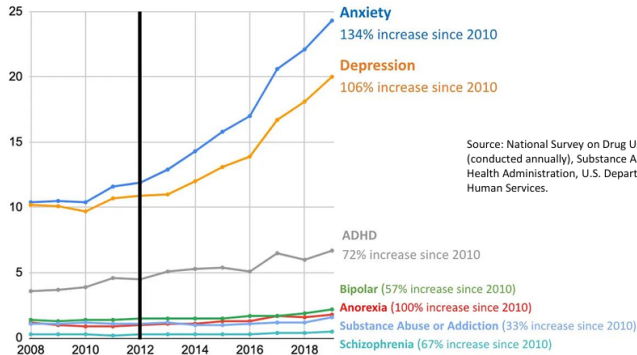
Depression, anxiety, and suicide are all rising among young people in some countries - roughly coinciding with the rise of social media use.

% US Teens with Major Depression



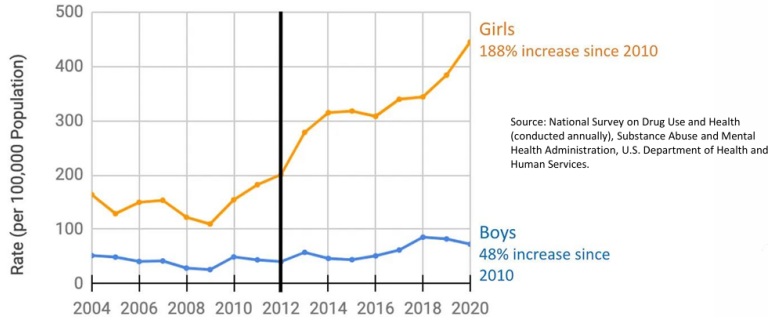
Impact on Young People: College Students

% of US Undergraduates Diagnosed with a Mental Illness



Impact on Young People: Self-Harm Hospitalizations

US Teens Admitted to Hospitals for Nonfatal Self-harm (Ages 10-14)



Alternative Interpretations of the Data

- Heavy social media use may be the **result** of depression rather than its cause
- Rising depression may reflect more frequent **reporting** and acknowledgment of psychological problems
- Rising depression may be real but have **other causes** that happen to coincide with social media use

The Ethical Question That Remains

We can't resolve the factual issues. But we **can** ask: assuming social media overuse causes depression, what should companies do about it?

Autonomy Issues and Children (I)

1 Denying Access

Denying access to YouTube (child or adult) is **no violation of autonomy**. Users cannot have an action plan of being granted access - only the platform can decide whether to grant it.

2 Children Are Not Fully Autonomous

Children often cannot form a coherent rationale for their behavior. In such cases, **parents can forbid** the behavior without violating autonomy.

Autonomy Issues and Children (II)

Children Are Nonetheless Agents

Children sometimes act autonomously - so injuring a child violates autonomy (as well as the utilitarian principle).

Why This Matters

Utilitarian benefits of allowing children online can **never outweigh** autonomy violations - children are far less capable than adults of giving informed consent to the risk of injury, so the chance of an autonomy violation is greater.

Conclusions (I)

Generalization Principle

- Removing clearly harmful content is generalizable
- Failure to remove it is also generalizable
- A carefully crafted policy of removing lies/falsehoods **may** be generalizable
- Flagging lies is generalizable

Autonomy Principle

Online sites can ethically operate only if one can rationally believe their moderated content will **never violate autonomy** without informed consent. Since children are agents, their autonomy must be protected.

Conclusions (II)

Utilitarian principle: a site should make a concerted effort to find an ethical content moderation policy, rather than shut down

It should **at least flag** false and misleading content, while citing reliable sources

Thank You